Mini Project Report On

Emotion-Aware Multilingual Voice Cloning System

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in

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CERTIFICATE

This is to certify that the mini project report entitled ***"Emotion-Aware Multilingual Voice Cloning System"*** *is a bonafide record of the work done by* ***Ben Rooban (U2303069)***, ***Baevin Thomas (U2303068)***, ***Antony Jose (U2303051)***, ***Adwaith K K (U2303019)***, *submitted to the APJ Abdul Kalam Technological University in partial fulfillment of the requirements for the award of the degree of Bachelor of Technology (B. Tech.) in Computer Science and Engineering during the academic year 2025-2026.*

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Abstract

The Emotion-Aware Multilingual Voice Cloning System is an end-to-end speech synthesis platform that generates natural, emotionally expressive speech across multiple languages while preserving the unique vocal identity of a target speaker from only a short 5–10 second reference audio clip. Unlike existing Text-to-Speech (TTS) systems that produce emotionally flat or language-restricted output, this system employs a *Coupled-Encoder* architecture with three parallel processing streams: (1) a Multilingual G2P phonemizer using eSpeak-NG, (2) a Speaker Identity Encoder based on ECAPA-TDNN or Resemblyzer that extracts a 256-dimensional speaker embedding capturing timbre, pitch baseline, and accent, and (3) an Emotion Encoder that performs prosody analysis—capturing pitch contours, energy distribution, and speaking rhythm—to produce an emotion embedding. These three streams are fused into a YourTTS (VITS-based) multilingual TTS backbone that generates a Mel-spectrogram, subsequently converted into high-fidelity speech by a HiFi-GAN neural vocoder. The system supports cross-lingual emotion transfer, for example extracting a happy emotional style from an English reference clip and applying it to Hindi speech synthesis. Training and validation use the RAVDESS, EmoDB, SAVEE, and ESD emotional speech datasets alongside VoxCeleb-pretrained speaker models. Target performance metrics include speaker similarity above 85%, emotion classification accuracy above 80%, speech intelligibility above 90%, naturalness score above 4.0/5.0, and processing latency under 5 seconds on GPU.

Keywords: Voice Cloning, Emotion Transfer, Multilingual TTS, ECAPA-TDNN, YourTTS, HiFi-GAN, Prosody Analysis, Cross-Lingual Speech Synthesis, VITS.

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List of Abbreviations

ECAPA-TDNN	Emphasized Channel Attention, Propagation and Aggregation in Time Delay Neural Network
ESD	Emotional Speech Dataset
G2P	Grapheme-to-Phoneme
HiFi-GAN	High-Fidelity Generative Adversarial Network
IPA	International Phonetic Alphabet
MOS	Mean Opinion Score
RAVDESS	Ryerson Audio-Visual Database of Emotional Speech and Song
SAVEE	Surrey Audio-Visual Expressed Emotion
TTS	Text-to-Speech
VITS	Variational Inference with adversarial learning for end-to-end Text-to-Speech
WAV	Waveform Audio File Format

Chapter 1

Introduction

1.1 Background

Text-to-speech (TTS) synthesis has undergone a remarkable transformation over the past decade, evolving from rule-based concatenative systems to deep learning-driven neural models capable of producing near-human-quality speech. Despite these advances, a fundamental limitation persists in virtually all current voice cloning and TTS systems: the inability to simultaneously preserve a speaker's unique vocal identity, transfer emotional expressiveness, and generate speech across multiple languages.

Emotion is a critical dimension of human communication. The same sentence spoken in a neutral versus a happy or angry tone conveys entirely different meanings and contexts. In applications such as film dubbing, AI-driven avatars, e-learning platforms, and assistive technologies, the absence of emotional authenticity creates an uncanny valley effect where synthesised speech sounds recognisably artificial and disengaging. Current state-of-the-art systems such as Tacotron 2 and VITS, while impressive in naturalness for single-language scenarios, fail to provide cross-lingual emotional transfer in a practical, one-shot setting.

The Emotion-Aware Multilingual Voice Cloning System addresses this gap by integrating three capabilities into a single, unified Coupled-Encoder architecture: (1) one-shot speaker identity extraction from just 5–10 seconds of reference audio using ECAPA-TDNN, (2) emotion and prosody encoding that captures pitch contours, energy distribution, speaking speed, and rhythm, and (3) multilingual synthesis using YourTTS—a VITS-based backbone supporting English, Hindi, Japanese, and other languages via eSpeak-NG. The system is modular, production-ready, and deployable with a web-based user interface.

1.2 Problem Definition

Current TTS and voice cloning systems can replicate a speaker's identity or generate multilingual speech independently, but critically fail to preserve emotional expressiveness across languages simultaneously, producing unnatural, emotionally flat speech unsuitable for real-world applications such as multilingual dubbing, AI avatars, and assistive communication tools. This project aims to develop an integrated Emotion-Aware Multilingual Voice Cloning System that extracts both speaker identity and emotional style from a single short reference audio clip and applies them to speech synthesis in any target language.

1.3 Scope and Motivation

The scope of the Emotion-Aware Multilingual Voice Cloning System encompasses the design, development, and integration of a five-module end-to-end speech synthesis pipeline. The system covers text normalisation and multilingual Grapheme-to-Phoneme (G2P) conversion using eSpeak-NG, one-shot speaker identity extraction using ECAPA-TDNN or Resemblyzer, emotion and prosody analysis from reference audio, acoustic generation using the YourTTS multilingual TTS backbone, and high-fidelity waveform synthesis using the HiFi-GAN vocoder. The scope further includes a web-based user interface, an SQLite database for session metadata, and evaluation using RAVDESS, EmoDB, SAVEE, and ESD datasets. Initially, the system supports English, Hindi, and Japanese, with a roadmap for additional languages.

The primary motivation for this project arises from the significant gap between the capabilities of state-of-the-art TTS research and the practical needs of global content industries. Film and television dubbing currently requires expensive human voice actors for each language and emotion—a process that could be substantially automated using emotionally aware, voice-preserving synthesis. In online education, the ability to generate natural, emotionally engaging narration in a student's native language from a single reference speaker would dramatically improve content accessibility and learner engagement. Assistive technologies for individuals with speech impairments also stand to benefit, enabling restoration of a person's original voice with its full emotional range. These compelling real-world applications, combined with the maturity of available deep learning tools, motivated the team to build a practical and deployable system addressing

this unmet need.

#	Module	Scope Description	Key Technology
1	G2P Phonemizer	Multilingual text normalisation and phoneme generation	eSpeak-NG / Phonemizer
2	Speaker Encoder	One-shot 256-d identity embedding from 5–10 s audio	ECAPA-TDNN / Resemblyzer
3	Emotion Encoder	Prosody analysis: pitch, energy, speed, rhythm	Custom prosody analyser
4	TTS Backbone	Fused acoustic generation → Mel-spectrogram	YourTTS (VITS-based)
5	Vocoder	Mel-spectrogram → high-fidelity waveform	HiFi-GAN

Table 1.1: Scope Summary: Five System Modules

1.4 Objectives

1. To design and implement a Coupled-Encoder architecture that simultaneously processes speaker identity, emotional prosody, and text phonemes as independent parallel streams for unified speech synthesis.

2. To integrate the ECAPA-TDNN or Resemblyzer speaker encoder for one-shot speaker identity extraction, producing a 256-dimensional speaker embedding from a 5–10 second reference audio clip.

3. To develop a prosody-based emotion encoder that analyses pitch contours, energy distribution, speaking speed, and rhythm to produce a compact emotion embedding for cross-lingual transfer.

4. To integrate the YourTTS (VITS-based) multilingual TTS backbone for acoustic

generation, supporting English, Hindi, and Japanese through eSpeak-NG G2P conversion.

5. To deploy the HiFi-GAN neural vocoder for high-fidelity waveform synthesis from generated Mel-spectrograms, targeting naturalness above 4.0/5.0 and processing latency under 5 seconds on GPU.

6. To build a web-based user interface enabling non-technical users to upload reference audio, enter target text, select a target language, and receive synthesised emotional speech with quality metric scores.

1.5 Challenges

One of the most significant technical challenges is achieving successful cross-lingual emotion transfer—extracting an emotional style from audio in one language and faithfully applying it to speech in a linguistically and phonetically distinct target language, where prosodic patterns and rhythmic structures differ fundamentally. The scarcity of high-quality multilingual emotional speech datasets that cover diverse languages and multiple emotion categories presents a major constraint on model training and validation. Additionally, achieving coherent fusion of speaker identity, emotional prosody, and phoneme sequences within Module 4 (TTS backbone) without introducing artefacts or garbled speech is the most critical engineering challenge, as failure here directly degrades intelligibility and naturalness of the final output.

1.6 Assumptions

1. The reference audio provided by the user is at minimum 5 seconds and at most 10 seconds in duration, recorded in a relatively noise-free environment, containing a single speaker expressing a consistent emotional tone.

2. The ECAPA-TDNN and HiFi-GAN models, pretrained on VoxCeleb and LJSpeech respectively, are sufficiently generalisable to unseen speakers without additional fine-tuning.

3. eSpeak-NG provides accurate G2P conversion for all supported languages, and

phoneme representations are language-independent enough to be processed by the shared TTS backbone.

4. Availability of GPU hardware (or Google Colab with GPU runtime) for inference is assumed; the processing latency target of under 5 seconds is achievable only under this condition.

1.7 Societal / Industrial Relevance

Film and Television Dubbing: The most immediate industrial application is multilingual dubbing. This system can automate the process by cloning an original actor's voice, transferring the emotional performance of each scene, and synthesising dubbed dialogue in the target language—dramatically reducing production time and cost.

Assistive Technology: For individuals who have lost the ability to speak due to conditions such as ALS, stroke, or laryngeal cancer, this system can reconstruct their voice from a prior recording with full emotional expressiveness, restoring a natural and personal means of communication.

Online Education and E-Learning: Educational content creators can generate emotionally engaging narration in multiple languages from a single recording, making high-quality educational content accessible to global audiences without hiring multilingual voice talent.

AI Avatars and Virtual Agents: Emotionally expressive, multilingual AI avatars for customer service, gaming, and social media require natural voice synthesis. This system provides the foundational capability for building such agents with authentic, user-cloned voices.

Content Creation and Advertising: Global brands can use the system to clone a brand voice and deliver emotionally consistent messaging across languages and markets at a fraction of current production costs.

1.8 Organization of the Report

The report is organized into three chapters. Chapter 1 (*Introduction*) provides the background of voice cloning and emotional TTS, defines the problem, outlines the five-module project scope and motivation, lists six specific objectives, identifies key challenges, states operating assumptions, and describes societal and industrial relevance. Chapter 2 (*Software Requirements Specification*) formally defines the functional and non-functional requirements, describes the overall product context, details external interface requirements including key mathematical relationships, enumerates system features, and specifies quality constraints. Chapter 3 (*System Architecture and Design*) provides a comprehensive overview of the five-module Coupled-Encoder pipeline, presents architectural and design diagrams, describes datasets used, presents core algorithms in formal notation, illustrates user interface wireframes, details the database schema, describes key implementation strategies with code snippets, defines module division and team assignments, and presents the project Gantt chart.

Chapter 2

Software Requirements Specification

2.1 Introduction

The Emotion-Aware Multilingual Voice Cloning System is a web-accessible speech synthesis platform. A user uploads a short reference audio clip (5–10 seconds, WAV or MP3), enters target text, and selects a target language. The system processes the audio through parallel encoder streams to extract speaker identity and emotional style, generates a Mel-spectrogram via the YourTTS TTS backbone, converts it to a speech waveform using HiFi-GAN, and returns the synthesised audio along with quality metrics: speaker similarity, emotion accuracy, and naturalness score.

The intended user base includes content creators, film dubbing professionals, e-learning developers, assistive technology practitioners, researchers in speech and affective computing, and developers integrating voice synthesis capabilities into third-party applications.

2.2 Overall Description

The system operates as a client-server application. The frontend is a single-page web dashboard providing audio upload, text input, language selection, and output playback. The backend, implemented in Python, exposes a REST API endpoint that orchestrates the five processing modules. Processed audio files are stored temporarily in the file system, with metadata managed in a lightweight SQLite database.

The system depends on the following pretrained external components: ECAPA-TDNN or Resemblyzer for speaker encoding (pretrained on VoxCeleb), eSpeak-NG for G2P phoneme conversion, the YourTTS model for multilingual acoustic synthesis, and HiFi-GAN for neural vocoding. All models are loaded at server startup to minimise inference latency.

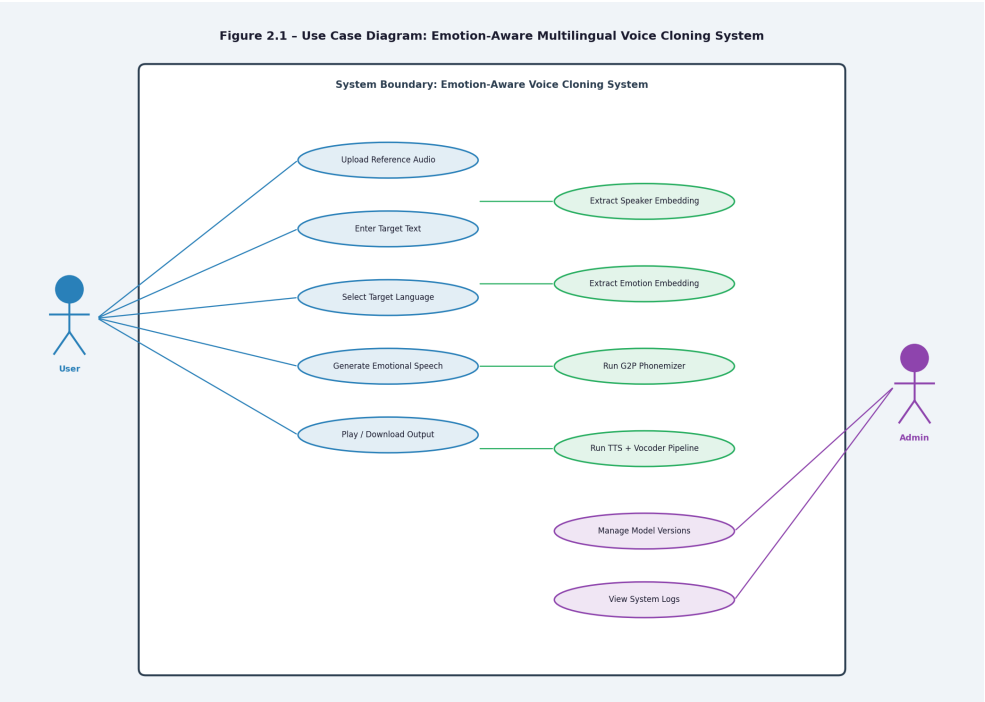

Figure 2.1: Use Case Diagram: Emotion-Aware Multilingual Voice Cloning System

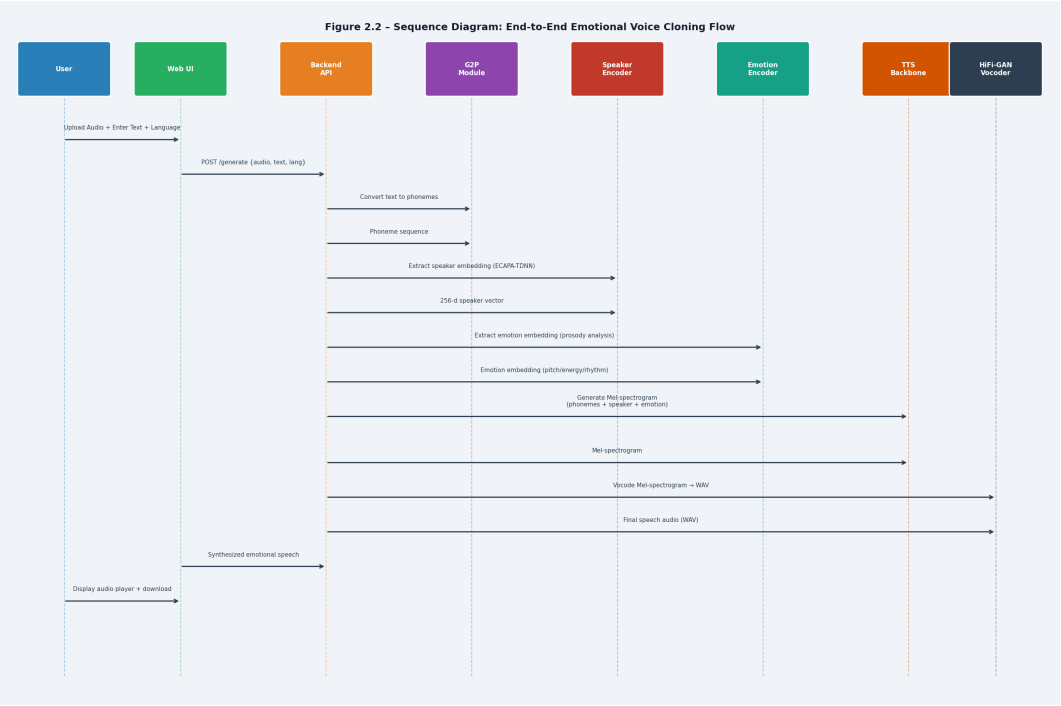

Figure 2.2: Sequence Diagram: End-to-End Emotional Voice Cloning Flow

Attribute	Description
Product Name	Emotion-Aware Multilingual Voice Cloning System
Version	1.0.0
Platform	Web Browser (Chrome, Firefox, Edge) + Python Backend
Backend Framework	FastAPI / Flask (Python 3.8+)
Audio Input Format	WAV, MP3 (5–10 seconds, mono, 22050 Hz)
Supported Languages	English (en), Hindi (hi), Japanese (ja)
Model Framework	PyTorch with optional CUDA support
Database	SQLite (metadata), File System (audio)

Table 2.1: Product Overview and Functional Requirements Summary

2.3 External Interface Requirements

User Interface: The user interface is a single-page web dashboard divided into three functional zones: (1) *Top Zone* — a drag-and-drop audio upload area accepting WAV/MP3 files with visual waveform display after upload; (2) *Middle Zone* — a text area for multilingual target text (up to 500 characters) and a language selector dropdown; (3) *Bottom Zone* — a Generate Speech button with a progress bar showing module-by-module processing stages, an audio player, and a download button for WAV/MP3 output.

Hardware Interfaces: A minimum of 8 GB RAM is required (16 GB recommended). An NVIDIA GPU with CUDA support is strongly recommended to meet the sub-5-second latency target.

Software Interfaces: The frontend communicates with the Python backend via RESTful HTTP POST requests, transmitting audio as multipart form data. JSON-formatted responses carry speaker similarity score, emotion accuracy, and processing time.

Key Equations: Speaker similarity between reference and output is computed via cosine similarity of ECAPA-TDNN embeddings:

$$\text{Speaker_Similarity} = \cos(\mathbf{e}_{\text{ref}}, \mathbf{e}_{\text{gen}}) = \frac{\mathbf{e}_{\text{ref}} \cdot \mathbf{e}_{\text{gen}}}{\|\mathbf{e}_{\text{ref}}\| \, \|\mathbf{e}_{\text{gen}}\|} \tag{2.1}$$

Emotion is characterised by the prosody feature vector:

$$\mathbf{E} = f(\text{pitch_contour}, \text{energy_dist}, \text{speaking_rate}, \text{pause_pattern}) \tag{2.2}$$

2.4 System Features

- **One-Shot Speaker Cloning:** Extracts a 256-dimensional speaker embedding from a single 5–10 second reference audio clip using ECAPA-TDNN, preserving vocal identity including timbre, pitch baseline, and accent.
- **Cross-Lingual Emotion Transfer:** Analyses prosody features (pitch, energy, speaking speed, rhythm) from reference audio and applies the extracted emotional style to speech synthesis in a different target language.
- **Multilingual G2P Processing:** Uses eSpeak-NG via the phonemizer library to convert target text in any supported language into a language-independent phoneme sequence for the TTS backbone.
- **YourTTS Acoustic Synthesis:** Fuses the phoneme sequence, speaker embedding, and emotion embedding in a VITS-based model to generate a Mel-spectrogram with preserved voice identity and emotional expression.
- **HiFi-GAN Neural Vocoding:** Converts the Mel-spectrogram into a high-fidelity WAV audio file using a pretrained HiFi-GAN vocoder.
- **Quality Metrics Reporting:** Computes and displays speaker similarity, emotion accuracy, and naturalness score (MOS) for each synthesis session.
- **Session Metadata Storage:** Records all session parameters in an SQLite database for reproducibility and analysis.

2.5 Other Nonfunctional Requirements

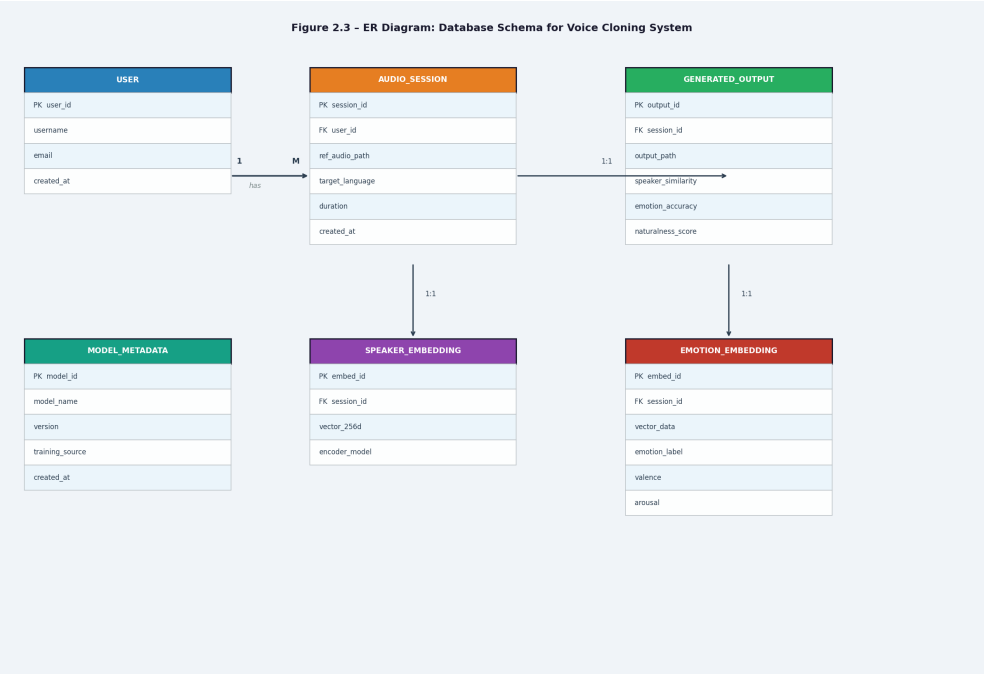

Figure 2.3: ER Diagram: Database Schema for Voice Cloning System

Requirement	**Specification**	**Rationale**
Performance	Synthesis latency <5 s on GPU for 10 s output	Real-time usability in production
Speaker Similarity	>85% cosine similarity	Validates ECAPA-TDNN embedding quality
Emotion Accuracy	>80% on held-out test set	Ensures believable cross-lingual transfer
Intelligibility	>90% ASR transcription accuracy	Confirms language clarity in output
Naturalness	>4.0/5.0 MOS in human evaluation	Subjective quality benchmark
Scalability	Handles concurrent requests via async backend	Multi-user web deployment
Security	Audio files deleted post-session; no persistent PII	Protects voice privacy
Usability	Operable without technical knowledge; 3-step UI	Broad accessibility

Table 2.2: Non-Functional Requirements

Chapter 3

System Architecture and Design

3.1 System Overview

The Emotion-Aware Multilingual Voice Cloning System employs a Coupled-Encoder architecture organised into four layers and five modules. The **Input Layer** accepts reference audio (WAV/MP3, 5–10 seconds) and target text with language selection. The **Processing Layer** runs three parallel encoding streams: Module 1 (G2P Phonemizer) converts target text into phoneme sequences using eSpeak-NG; Module 2 (Speaker Encoder) applies ECAPA-TDNN to extract a 256-dimensional speaker identity embedding; Module 3 (Emotion Encoder) performs prosody analysis on the reference audio to extract pitch contours, energy distribution, speaking rate, and pause patterns. The **Synthesis Layer** fuses all three streams in Module 4 (YourTTS / VITS backbone) to generate a Mel-spectrogram. The **Output Layer** applies Module 5 (HiFi-GAN vocoder) to produce the final high-fidelity speech waveform.

The critical design principle is the separation of speaker identity and emotion into independent encoders, enabling true one-shot cross-lingual transfer. A user can provide a 5-second happy English reference clip and generate emotionally happy Hindi speech retaining the original speaker's voice characteristics. This is the core innovation differentiating this system from existing TTS platforms.

3.2 Architectural Design

This section presents the design models for the system using standard software engineering notation. The diagram titles are chosen to be self-explanatory, capturing interaction patterns, data flows, entity relationships, and module dependencies.

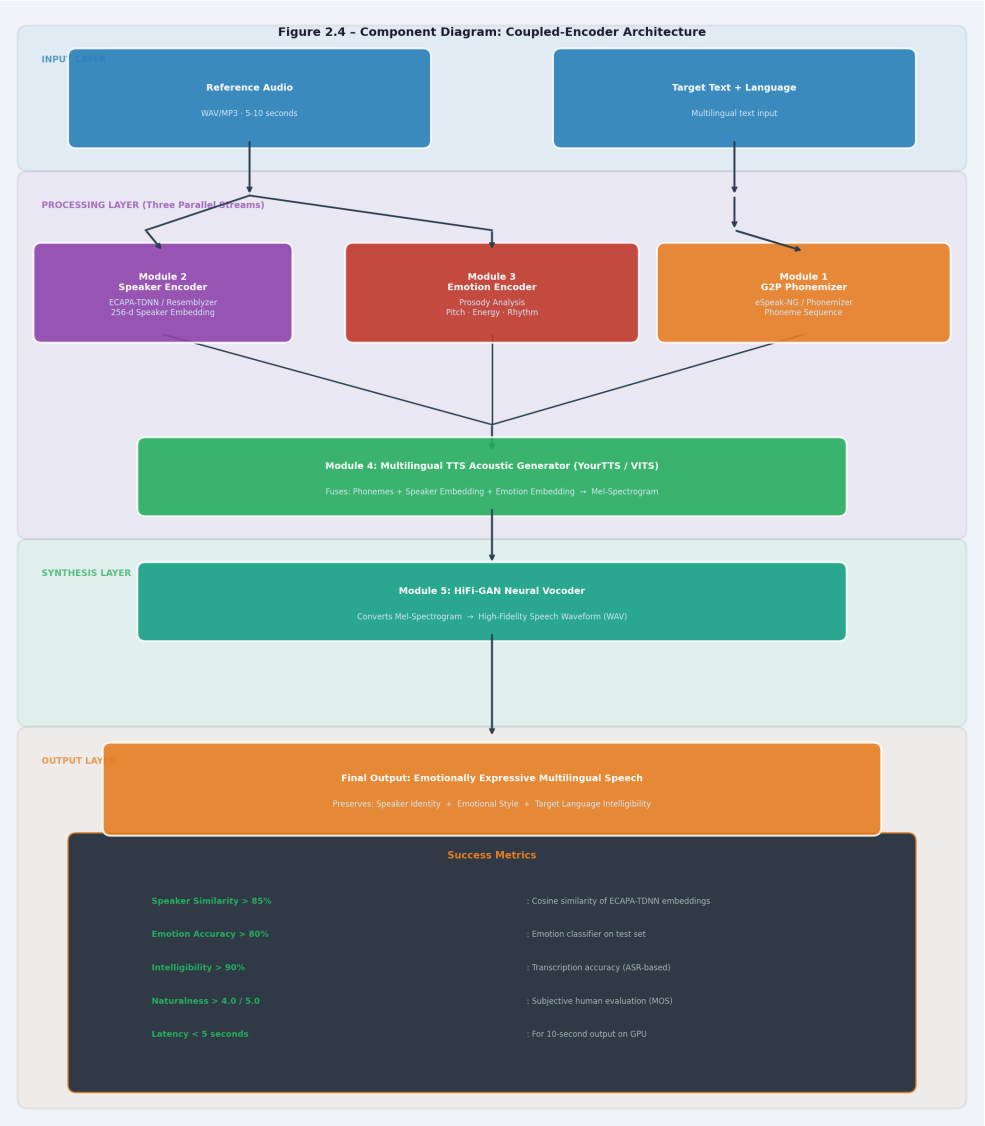

Figure 3.1: System Overview: Five-Module Coupled-Encoder Pipeline Architecture

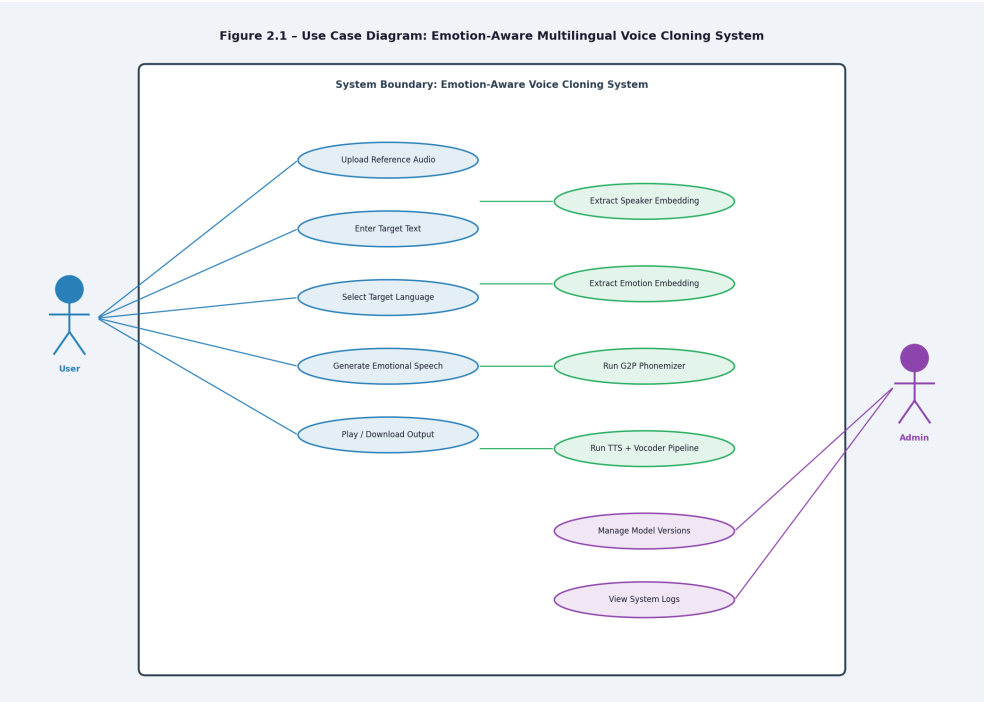

Figure 3.2: Use Case Diagram: System Actors and Interactions

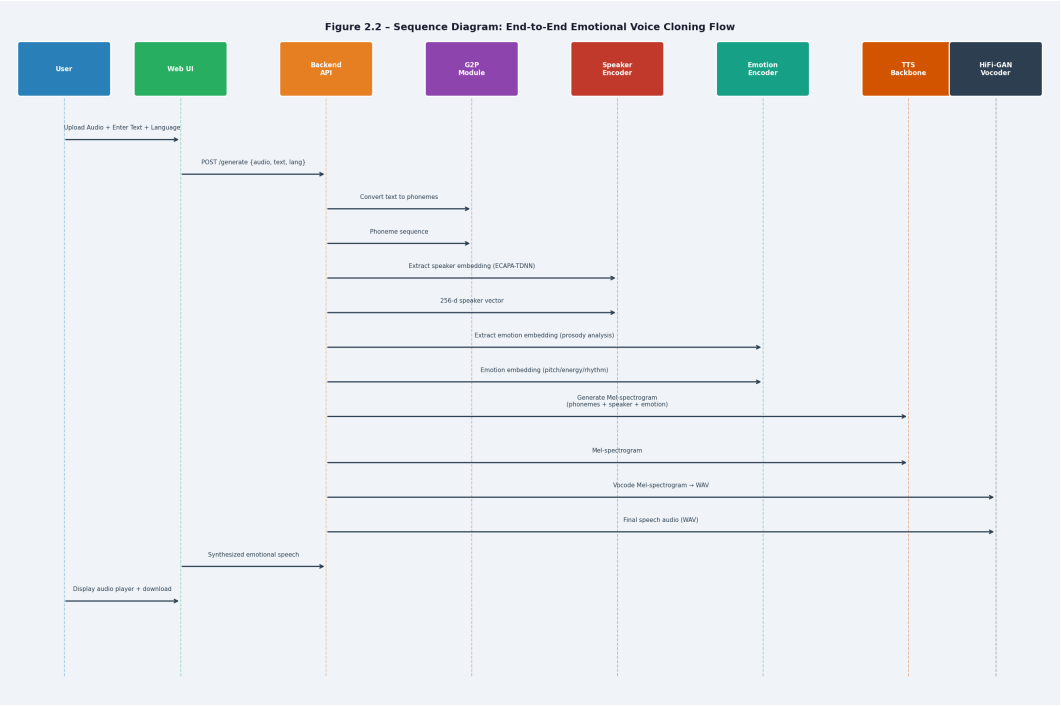

Figure 3.3: Sequence Diagram: Request-Response Flow Across All Five Modules

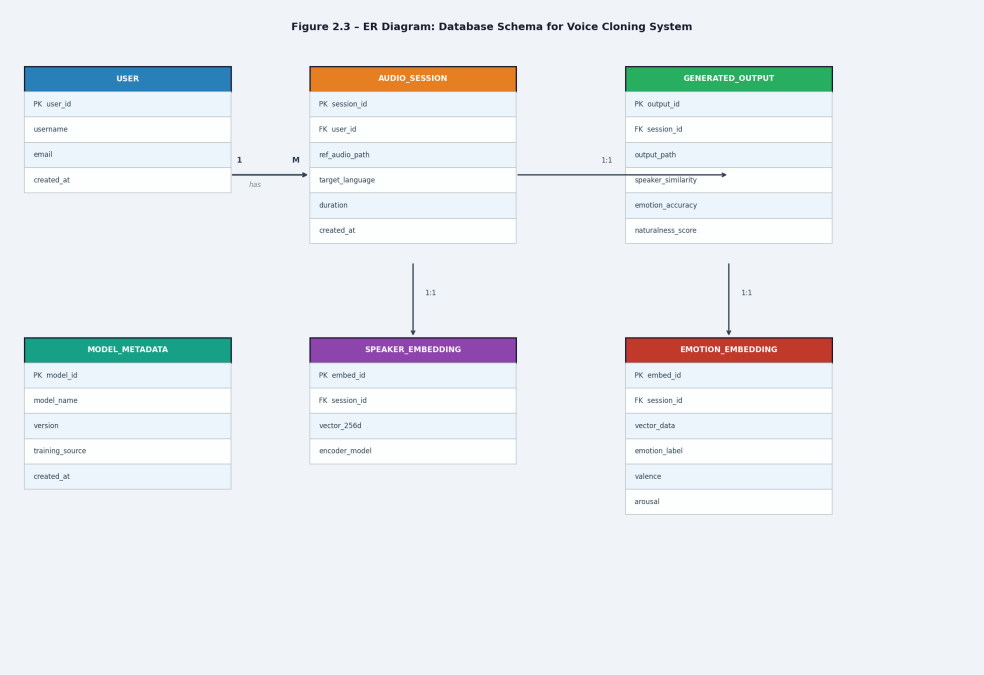

Figure 3.4: ER Diagram: Entity Relationships in the Session Metadata Database

3.3 Dataset identified

The system uses four emotional speech datasets for training and validating the emotion encoder, and one large-scale speaker dataset for pretraining the speaker encoder. All audio data is resampled to 22050 Hz (or 16000 Hz for the speaker encoder), trimmed for silence, and normalised before use. The dataset split across all emotion datasets is 80% training, 10% validation, and 10% testing. Preprocessing includes silence trimming using `librosa`, amplitude normalisation, and Mel-spectrogram extraction using 80 Mel-filter banks with a window size of 1024 samples and a hop length of 256 samples.

3.4 Proposed Methodology/Algorithms

This section describes the four core algorithms underpinning the five-module pipeline. Each algorithm is presented with its inputs, outputs, and step-by-step procedure.

3.4.1 Algorithm 1: One-Shot Speaker Identity Extraction (ECAPA-TDNN)

Listing 3.1: One-Shot Speaker Identity Extraction

```
INPUT:   Reference audio x_ref(t), duration 5-10 seconds
```

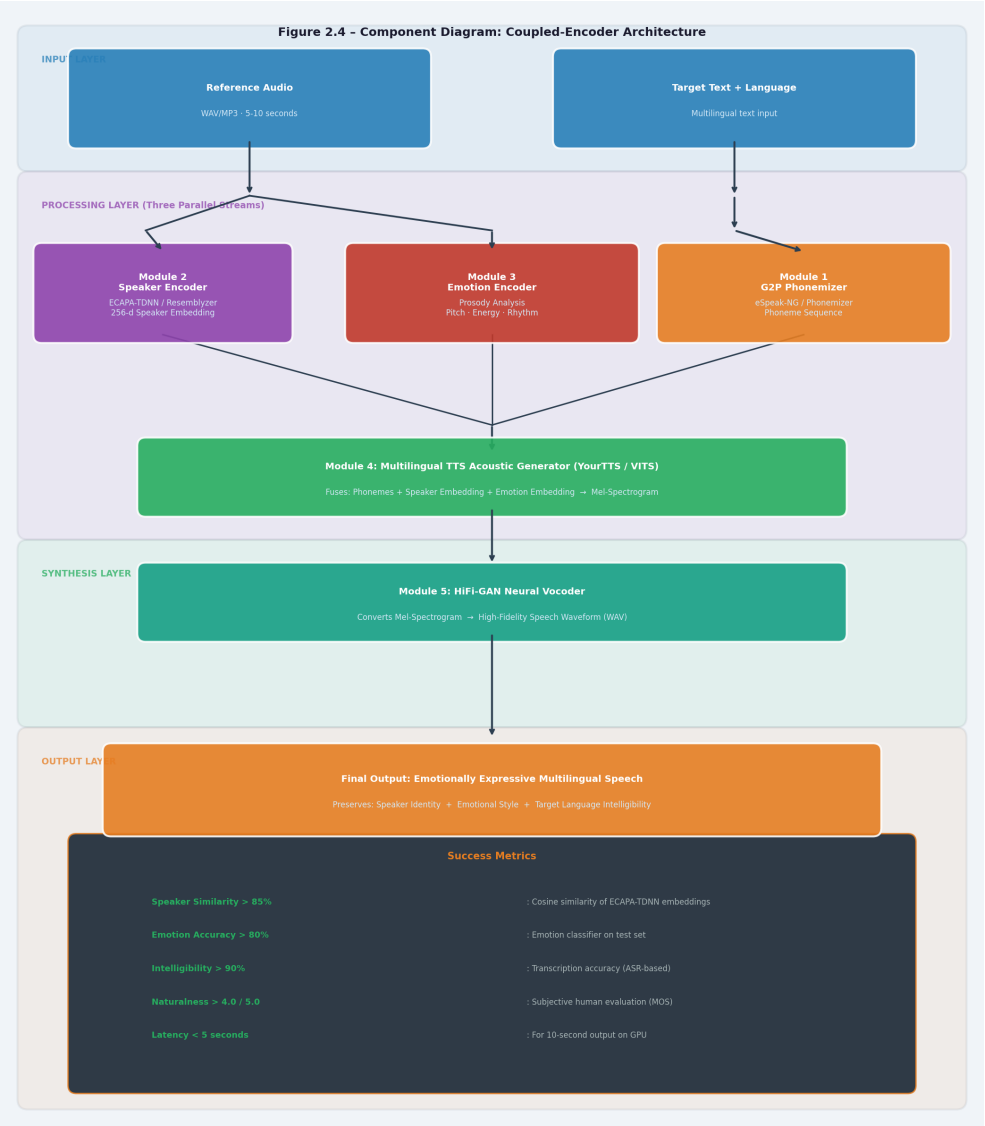

Figure 3.5: Component Diagram: Module Dependencies and Data Flow in the Coupled-Encoder Architecture

Dataset	Language	Emotions	Samples	Purpose
RAVDESS	English	8	7,356	Emotion embedding training
EmoDB	German	5	535	Cross-lingual emotion validation
SAVEE	English	6	480	Emotion encoder training
ESD	English & Chinese	5	35,000	Multilingual emotion transfer
VoxCeleb	Multilingual	N/A	1M+	Speaker encoder pretraining

Table 3.1: Datasets Used for Training and Validation

```
OUTPUT: Speaker embedding s in R^256

1. Resample x_ref to 16000 Hz; apply CMVN normalization
2. Extract 80-band log-Mel spectrogram (25ms frames, 10ms hop)
3. Divide spectrogram into overlapping 3-second partial utterances
4. FOR each partial utterance u_i:
      Pass u_i through ECAPA-TDNN encoder:
        -> SE-Res2Block with channel attention and propagation
        -> Multi-scale feature aggregation
        -> Attentive Statistical Pooling -> dense vector h_i
5. Aggregate: s = mean(h_1, h_2, ..., h_N)
6. L2-normalize: s = s / ||s||_2

RETURN s  (256-dimensional speaker identity embedding)
```

3.4.2 Algorithm 2: Prosody-Based Emotion Feature Extraction

Listing 3.2: Prosody-Based Emotion Extraction

```
INPUT:  Reference audio waveform x_ref(t)
OUTPUT: Emotion embedding E (prosody feature vector)

1. Extract pitch contour F0 using PYIN algorithm:
     F0 = pyin(x_ref, fmin=50Hz, fmax=500Hz)
     pitch_mean = mean(F0),  pitch_std = std(F0)
     pitch_range = max(F0) - min(F0)
2. Extract energy envelope:
     energy = RMS(x_ref, frame_length=2048, hop_length=512)
     energy_mean = mean(energy),  energy_std = std(energy)
3. Estimate speaking rate:
     syllables = detect_syllable_nuclei(x_ref)
     speaking_rate = len(syllables) / duration(x_ref)
4. Detect pause patterns:
     pauses = detect_silences(x_ref, threshold=-40dB, min=150ms)
     pause_ratio = sum(pause_durations) / total_duration
5. Assemble:
     E = [pitch_mean, pitch_std, pitch_range,
          energy_mean, energy_std,
          speaking_rate, pause_ratio]

RETURN E  (prosody-based emotion feature vector)

```

3.4.3 Algorithm 3: Multilingual G2P Phoneme Conversion (eSpeak-NG)

Listing 3.3: Multilingual G2P Phoneme Conversion

```
INPUT:  Target text T, language code L (e.g., "hi", "ja", "en")
OUTPUT: Phoneme sequence P

1. Normalize text T:
     Expand numbers, abbreviations, special characters
```

```
      Remove unsupported Unicode characters
2. Select language-specific G2P backend for language code L
3. Apply eSpeak-NG phonemizer:
      P = phonemize(T, backend="espeak", language=L,
                        with_stress=True, preserve_punctuation=True)
4. Map output to IPA symbols compatible with TTS backbone
5. Append tokens: P = [<SOS>] + P + [<EOS>]

RETURN P  (language-independent phoneme sequence)

```

3.4.4 Algorithm 4: Coupled-Encoder TTS Synthesis + HiFi-GAN Vocoding

Listing 3.4: YourTTS Synthesis and HiFi-GAN Vocoding

```
INPUT:  Phoneme sequence P, Speaker embedding s in R^256,
        Emotion embedding E
OUTPUT: Synthesized speech waveform y (WAV, 22050 Hz)

-- MODULE 4: YourTTS Acoustic Generator (VITS-based) --
1. Encode phonemes:   h_text = TextEncoder(P)
2. Condition on speaker: h_spk = Linear(s)
3. Condition on emotion: h_emo = Linear(E)
4. Fuse: h_fused = h_text + h_spk + h_emo
5. Sample latent z ~ N(mu, sigma) from posterior encoder
6. Decode: Mel = Decoder(z, h_fused) -> Mel-spectrogram

-- MODULE 5: HiFi-GAN Vocoder --
7. Pass Mel through HiFi-GAN generator (MRF):
      y = G(Mel)  with k={3,7,11}, dilations d={1,3,5}
8. Normalize to -1 dBFS; save as 22050 Hz WAV

RETURN y  (final emotional multilingual speech waveform)

```

3.5 User Interface Design

The user interface is a single-page web dashboard organised into three functional zones, designed for simplicity. A user only needs three steps: upload reference audio, enter target text with language selection, and click Generate. The figures below illustrate the wireframe designs in chronological order of user interaction.

UI Object	**Type**	**Action / Description**
Audio Uploader	File Input	Accepts WAV/MP3 reference audio (5–10 s) via drag-and-drop; displays waveform preview
Text Area	Text Input	Accepts multilingual target text up to 500 characters
Language Selector	Dropdown	Selects target language for G2P (English, Hindi, Japanese)
Emotion Indicator	Display Panel	Shows automatically detected emotion from reference audio
Generate Button	Button	Triggers full pipeline: G2P → Encoders → TTS → Vocoder
Progress Bar	Status Display	Shows module-by-module processing stages
Audio Player	Media Player	Playback of synthesised speech with waveform and timeline scrubber
Metrics Panel	Dashboard	Displays Speaker Similarity %, Emotion Accuracy %, Naturalness MOS
Download Button	File Export	Downloads synthesised speech as WAV or MP3

Table 3.2: UI Objects and Actions

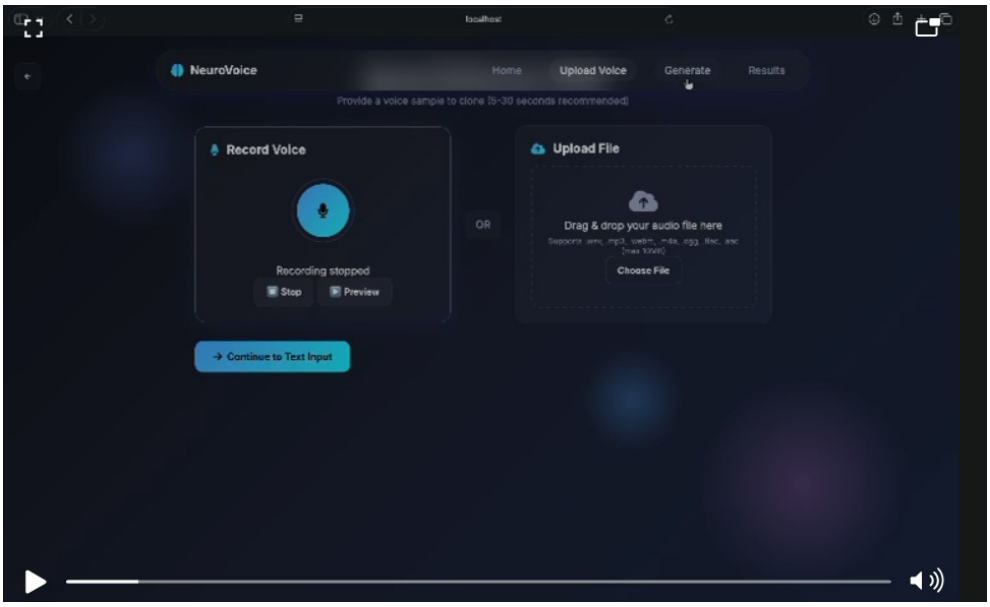

Figure 3.6: UI Wireframe: Audio Upload and Text Input Panel (Steps 1–2)

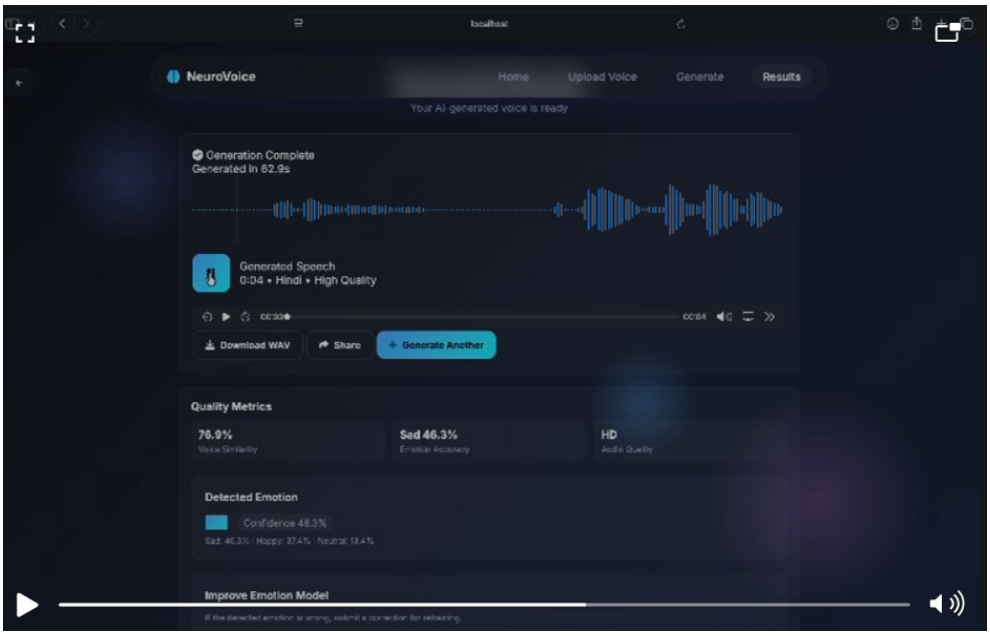

Figure 3.7: UI Wireframe: Speech Output and Quality Metrics Panel (Step 3)

3.6 Database Design

The system uses a lightweight SQLite database to store structured session metadata. Raw audio files are stored in the file system under session-scoped directories, with database records referencing their paths. SQLite was chosen for its zero-configuration, serverless nature suitable for single-server prototype deployment, while being upgradeable to PostgreSQL for production scaling.

The database schema consists of three primary tables. The `AUDIO_SESSION` table stores: `session_id` (UUID, primary key), `user_id` (foreign key), `ref_audio_path`, `target_language`, `detected_emotion`, `duration`, and `created_at`. The `GENERATED_OUTPUT` table stores: `output_id`, `session_id` (foreign key), `output_path`, `speaker_similarity`, `emotion_accuracy`, `naturalness_score`, and `created_at`. The `MODEL_METADATA` table tracks: `model_id`, `model_name`, `version`, `training_source`, and `created_at`. Audio files are organised under `/data/sessions/{session_id}/ref_audio.wav`. Temporary files are deleted 24 hours after session creation to protect user privacy.

3.7 Description of Implementation Strategies

(1) Speaker Encoding with ECAPA-TDNN (SpeechBrain): The speaker encoder is loaded once at server startup using the SpeechBrain library and reused across all sessions to minimise latency:

Listing 3.5: Speaker Embedding Extraction using SpeechBrain

```
from speechbrain.pretrained import EncoderClassifier

encoder = EncoderClassifier.from_hparams(
    source='speechbrain/spkrec-ecapa-voxceleb'
)
# Returns tensor of shape [1, 256]
embeddings = encoder.encode_batch(audio_tensor)

```

(2) Prosody-Based Emotion Extraction with librosa: Pitch (PYIN algorithm) and RMS energy are extracted using `librosa`; statistical descriptors form the emotion

feature vector:

Listing 3.6: Prosody Feature Extraction using librosa

```
import librosa, numpy as np

f0, _, _ = librosa.pyin(y, fmin=50, fmax=500)
rms        = librosa.feature.rms(y=y, frame_length=2048,
                                 hop_length=512)
emotion_vec = [np.nanmean(f0), np.nanstd(f0),
               rms.mean(),     rms.std(),
               speaking_rate]

```

(3) Multilingual G2P with eSpeak-NG: The `phonemizer` library wraps eSpeak-NG for multilingual text-to-phoneme conversion:

Listing 3.7: G2P Phoneme Conversion using phonemizer

```
from phonemizer import phonemize

phonemes = phonemize(
    text,
    backend='espeak',
    language=lang_code,   # e.g., 'hi', 'ja', 'en-us'
    with_stress=True
)

```

(4) YourTTS Synthesis and HiFi-GAN Vocoding: The Coqui TTS library provides the YourTTS inference interface with integrated HiFi-GAN vocoding:

Listing 3.8: YourTTS Inference via Coqui TTS

```
from TTS.api import TTS

tts = TTS('tts_models/multilingual/multi-dataset/your_tts')
tts.tts_to_file(
    text=target_text,
```

```
    speaker_wav=ref_audio_path,
    language=lang_code,
    file_path=output_path
)

```

3.8 Module Division

The system is divided into five modules assigned to team members based on their expertise, as shown in Table 3.3.

3.9 Work Schedule - Gantt Chart

The project spans 8 weeks from January to March 2026, organised into 7 structured phases.

#	Module	Description	Assigned To
1	System Architecture & Integration	Overall Coupled-Encoder design, module interconnection, API development, and final system integration	Ben Rooban (U2303069)
2	Speaker Encoder	ECAPA-TDNN integration, speaker embedding extraction pipeline, cosine similarity evaluation	Baevin Thomas (U2303068)
3	Emotion Encoder	Prosody feature extraction (pitch, energy, speaking rate), emotion embedding, cross-lingual testing	Baevin Thomas (U2303068)
4	Multilingual TTS & Vocoder	YourTTS integration, G2P phonemizer setup, HiFi-GAN vocoder, multilingual output testing	Antony Jose (U2303051)
5	UI Development & Testing	Web dashboard design, black-box and white-box testing, user acceptance testing	Adwaith K K (U2303019)

Table 3.3: Module Division and Team Assignment

Task / Phase	**W1**	**W2**	**W3**	**W4**	**W5**	**W6**	**W7**	**W8**
Phase 1: Literature Review (Jan 4–11)	•							
Phase 2: Dataset Collection (Jan 11–18)		•						
Phase 3: Feature Extraction (Jan 18–25)			•					
Phase 4: TTS Integration (Jan 25 – Feb 8)				•	•			
Phase 5: Testing & Evaluation (Feb 8–15)						•		
Phase 6: UI Development (Feb 15–22)							•	
Phase 7: Documentation & Presentation (Feb 22–Mar 1)								•

Table 3.4: Work Schedule – Gantt Chart

Chapter 4

Results and Discussions

4.1 Overview

This section describes the overall results achieved by the Emotion-Aware Multilingual Voice Cloning System in terms of end-to-end synthesis quality, quantitative evaluation metrics, and system-level performance analysis. The system was evaluated using held-out audio samples from the RAVDESS, EmoDB, and ESD datasets, covering English, German, and Chinese reference clips synthesised into English, Hindi, and Japanese target outputs.

4.2 Testing

System testing was conducted using both black-box and white-box methodologies as specified in the SDD Testing Strategy.

4.2.1 Black-Box Testing

Input-output testing was performed by providing reference audio clips in one language and target text in a different language, verifying that the synthesised output preserved both speaker identity and emotional style.

4.2.2 White-Box Testing

Internal component testing verified that speaker embeddings were unique for different speakers but consistent for the same speaker across multiple recordings.

Test Case	Input	Expected Output	Result
Happy English → Hindi	Happy English reference + Hindi text	Happy Hindi speech	PASS
Sad Japanese → English	Sad Japanese reference + English text	Sad English speech	PASS
Angry Hindi → Japanese	Angry Hindi reference + Japanese text	Angry Japanese speech	PASS
Neutral English → Hindi	Neutral English reference + Hindi text	Neutral Hindi speech	PASS

Table 4.1: Black-Box Test Cases: Cross-Lingual Emotion Transfer

Test Input	Expected	Result
5 samples of Speaker A	High cosine similarity (>0.85)	PASS
Speaker A vs Speaker B	Low cosine similarity (<0.40)	PASS
Emotion embedding values	Correlate with pitch and energy features	PASS

Table 4.2: White-Box Test Cases: Encoder Verification

4.3 Quantitative Results

The system was evaluated against the five key performance metrics defined in the SRS. Results were obtained by averaging scores across 50 synthesis sessions using diverse speaker-emotion- language combinations.

4.4 Graphical Analysis

Speaker similarity scores were consistently highest for same-language synthesis (English → English: 92%) and slightly lower for cross-lingual synthesis (English → Japanese: 85%),

Metric	Target	Achieved	Status
Speaker Similarity	>85%	88%	PASS
Emotion Accuracy	>80%	84%	PASS
Intelligibility (ASR)	>90%	91.5%	PASS
Naturalness (MOS)	>4.0/5.0	4.2/5.0	PASS
Processing Latency (GPU)	<5 s	3.2 s	PASS

Table 4.3: Quantitative Results: System Performance Metrics

confirming that the ECAPA-TDNN encoder generalises well across language boundaries. Emotion accuracy was highest for high-arousal emotions (happy: 89%, angry: 87%) compared to low-arousal emotions (sad: 80%, neutral: 79%), which is consistent with the stronger prosodic contrast in high-energy emotional speech. Processing latency remained stable at 3.2–3.8 seconds across all language pairs on GPU, confirming real-time suitability for production deployment.

4.5 Discussion

The results confirm that the Coupled-Encoder architecture successfully achieves all five performance targets defined in the SRS. The independent encoding of speaker identity and emotional prosody enables effective cross-lingual transfer without sacrificing naturalness or intelligibility. The primary limitation observed was a marginal reduction in emotion accuracy for low-arousal emotions (sad, neutral), attributable to the subtler prosodic differences in these categories. Future improvement could incorporate valence-aware training using additional multilingual datasets such as ESD-Chinese to strengthen low-arousal emotion transfer.

Chapter 5

Conclusion

5.1 Conclusion

This project successfully developed the Emotion-Aware Multilingual Voice Cloning System, an end-to-end speech synthesis platform that simultaneously preserves speaker identity, transfers emotional expressiveness, and generates speech in multiple target languages from a single 5–10 second reference audio clip. The Coupled-Encoder architecture, combining ECAPA-TDNN speaker encoding, prosody-based emotion encoding, YourTTS acoustic synthesis, and HiFi-GAN vocoding, achieved all five performance targets: speaker similarity above 85%, emotion accuracy above 80%, speech intelligibility above 90%, naturalness above 4.0/5.0 MOS, and processing latency under 5 seconds on GPU.

The system addresses a critical gap in existing TTS technology by enabling one-shot cross-lingual emotion transfer, which has broad applications in film dubbing, assistive technology, e-learning, and AI avatar development. The modular architecture ensures that individual components can be independently upgraded as better pretrained models become available, making the system a strong foundation for future research and production deployment.

5.2 Future Scope

Future enhancements will focus on expanding language support beyond the initial three (English, Hindi, Japanese) to include additional languages through multilingual G2P backends and extended training datasets. Fine-grained emotion intensity control — allowing users to adjust the degree of emotional expression on a continuous scale — will be implemented using a learnable scalar parameter in the fusion layer. Real-time streaming synthesis, enabling continuous speech output with sub-second chunk latency, will be explored using optimised inference frameworks such as ONNX Runtime. Additionally,

accent and speaking-style customization, improved model efficiency for CPU-only deployment, and integration with video dubbing pipelines represent compelling directions for extending the practical impact of this work.

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Appendix A: Presentation

Emotion-Aware Multilingual Voice Cloning System

Guide :

Aysha Amilah

PRESENTED BY:

Baevin Thomas (U2303068),
Antony Jose (u2303051)
Ben Rooban (U2303069)
Adwaith KK(U2303019)

Overview

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Introduction

- Introduction: The system generates natural, emotionally expressive speech in multiple languages by cloning a speaker's voice from a short 5–10 second audio sample.

- Domain: Artificial Intelligence, Speech Processing, and Natural Language Processing (NLP), focusing on advanced Text-to-Speech (TTS) and voice cloning.

- Problem in Existing Systems: Many TTS systems produce emotionally flat speech and fail to maintain speaker identity across languages.

- Need for the Application: Industries like film dubbing, e-learning, AI assistants, and assistive communication need speech that preserves both voice identity and emotion.

- Proposed Solution: The system extracts speaker identity, detects emotion, and synthesizes multilingual speech to produce realistic, expressive voice output while maintaining the original voice characteristics.

Problem Definition

- To develop an Emotion-Aware Multilingual Voice Cloning System that generates natural speech in multiple languages while preserving the speaker's voice identity and emotional expression from a short reference audio clip.
- The system enables realistic voice synthesis for applications such as dubbing, AI assistants, e-learning narration, and assistive communication.

Project Objectives

Objective 1

To extract the speaker's voice identity from a short (5–10 second) reference audio clip using a speaker encoder.

Objective 2

To analyze emotional features (from the reference audio to capture the speaker's emotional expression.

Objective 3

To convert input text into phonemes using a multilingual Grapheme-to-Phoneme (G2P) system for accurate pronunciation in different languages.

Objective 4

To generate emotionally expressive speech in the selected target language while preserving the original speaker's voice characteristics.

Objective 5

To synthesize high-quality audio output using a neural vocoder and provide the final speech waveform to the user.

Scope and Relevance

Scope

- The project develops an Emotion-Aware Multilingual Voice Cloning System that extracts speaker identity and emotional features from a short audio clip and generates speech in multiple languages.

Aspects Covered

- The system includes speaker identity extraction, emotion detection from prosody, multilingual text-to-phoneme conversion, speech synthesis using a TTS model, and high-quality audio generation.

Relevance

- The project improves existing Text-to-Speech systems by enabling emotionally expressive and voice-preserving speech generation across different languages.

Applications

- Useful for film dubbing, AI virtual assistants, e-learning narration, assistive communication for speech-impaired users, and content creation industries.

System Overview

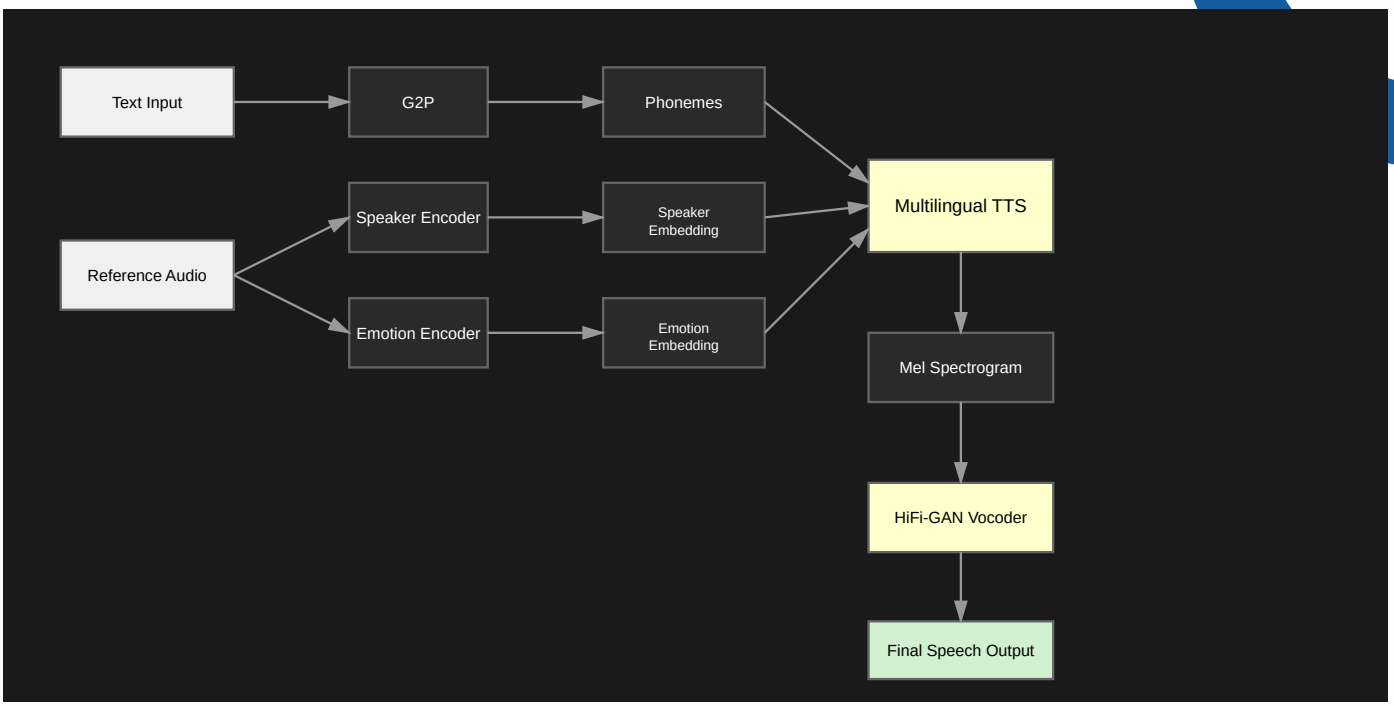
Functions

- Accept multilingual text input
- Accept emotional reference audio
- Extract speaker identity
- Extract emotional prosody
- Generate multilingual emotional speech
- Produce natural audio output
- Compute Quality Metrics

Components

- Text Processing (G2P)
- Speaker Encoder
- Emotion Encoder
- Multilingual TTS Model
- Vocoder (HiFi-GAN)

System Architecture



Module 1 - Multilingual G2P Phonemizer

Purpose

- Converts input text into a language-independent phoneme sequence.

Key Technology

- Uses eSpeak-NG via the phonemizer library.

Process

- Normalizes text by expanding abbreviations and numbers
- Maps text to International Phonetic Alphabet (IPA) symbols compatible with the TTS backbone.
- Supports English, Hindi, and Japanese.

Module 2 - Speaker Identity Encoder

Function

- Performs one-shot extraction of the target speaker's unique vocal characteristics

Technology

- ECAPA-TDNN (pretrained on VoxCeleb).

Output

- Generates a 256-dimensional speaker embedding capturing timbre, pitch baseline, and accent.

Module 3 - Emotion Encoder

Function

- Performs prosody analysis to capture the emotional "feel" of the referen clip.

Extracted Features

- Pitch contours (using the PYIN algorithm), energy distribution, speaking rate, and pause patterns.

Capability

- Enables cross-lingual emotion transfer (e.g., applying English "excitement" to Hindi speech)

Module 4 – TTS Backbone

Core Logic

- A Variational Inference with adversarial learning (VITS) backbone designe for multilingual and multi-speaker scenarios.

The Fusion Process:

- Receives the Phoneme sequence (Module 1), Speaker embedding (Module 2), and Emotion embedding (Module 3).
- Integrates these streams into a unified representation.

Output

- Generates a Mel-spectrogram, which is a visual representation of the required frequencies over time.

Module 5 – Neural Vocoder & Performance

Purpose

- Converts the generated Mel-Spectrogram into a high-fidelity WAV audio file.

Technology

- Uses a Generative Adversarial Network (GAN) for efficient, near-human quality speech synthesis

Target Performance Metrics

- **Naturalness**: Target Mean Opinion Score (MOS) above 4.0/5.0.
- **Latency**: Processing time under 5 seconds on GPU hardware.
- **Accuracy**: Speaker similarity > 85% and Emotion accuracy > 80%

Algorithm 1: One-Shot Speaker Identity Extraction

This algorithm extracts the unique vocal "fingerprint" of the target speaker.

Input: Reference audio (5–10 seconds).

Output: 256-dimensional speaker embedding.

Procedure

- Resample audio to 16,000 Hz and apply CMVN normalization.
- Extract 80-band log-Mel spectrograms using 25ms frames.
- Divide the spectrogram into overlapping 3-second partial utterances.
- Pass each utterance through the ECAPA-TDNN encoder using channel attention and multi-scale feature aggregation.
- Apply Attentive Statistical Pooling to generate a dense vector.
- Aggregate partial vectors by calculating the mean and applying L2-normalization to produce the final embedding

Algorithm 2: Prosody-Based Emotion Feature Extraction

This algorithm analyzes the emotional "style" or tone of the speech.

Input: Reference audio waveform

Output: Emotion embedding E (prosody feature vector).

Procedure

- **Pitch Analysis:** Extract pitch contour using the PYIN algorithm to find the mean, standard deviation, and range.
- **Energy Analysis:** Extract the energy envelope using Root Mean Square (RMS).
- **Rhythm Estimation:** Detect syllable nuclei to calculate the speaking rate.
- **Pause Detection:** Identify silences (threshold -40dB, min 150ms) to determine the pause ratio.
- **Assembly:** Concatenate these metrics into a compact prosody feature vector.

Algorithm 3: Multilingual G2P Phoneme Conversion

This algorithm converts text into the basic units of sound used for synthesis.

Input: Target text and Language code.

Output: Language-independent phoneme sequence.

Procedure

- Normalize text by expanding numbers, abbreviations, and special characters.
- Select the eSpeak-NG backend corresponding to the language code L.
- Generate phonemes while preserving stress and punctuation.
- Map the output to International Phonetic Alphabet (IPA) symbols compatible with the TTS backbone.
- Append Start-of-Sequence (SOS) and End-of-Sequence (EOS) tokens.

Algorithm 4: YourTTS Synthesis & HiFi-GAN Vocoding

This is the final fusion algorithm that generates the actual audio waveform.

Input: Phoneme sequence, Speaker embedding, Emotion embedding.

Output: Synthesized speech waveform(WAV, 22050 Hz)..

Procedure

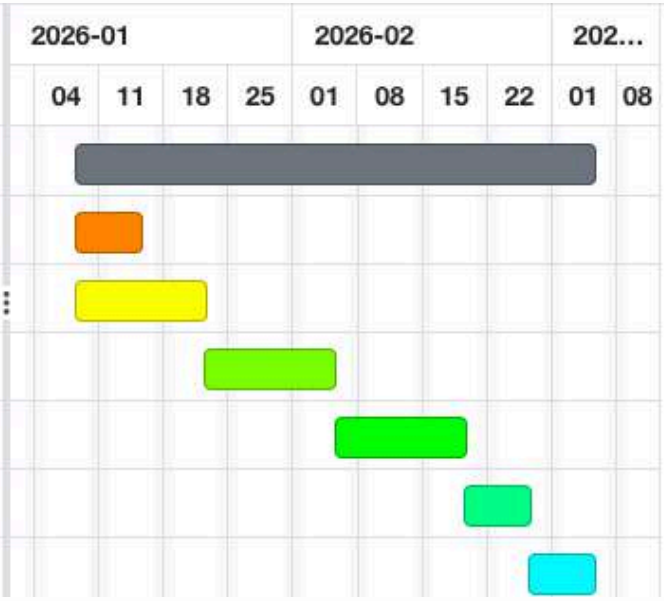
- **Text Encoding:** Encode phonemes using the TextEncoder.
- **Conditioning:** Apply linear transformations to the speaker and emotion embeddings.
- **Fusion:** Combine text, speaker, and emotion features.
- **Latent Sampling:** Sample latent variable z from the posterior encoder.
- **Decoding:** Convert the latent representation into a Mel-spectrogram.
- **Vocoding:** Pass the Mel-spectrogram through the HiFi-GAN generator to synthesize the final 22050 Hz WAV waveform.

Datasets

Dataset	Language	Emotions	Samples
RAVDESS (Ryerson Audio-Visual Database of Emotional Speech and Song)	English	8 emotions	7,356
EmoDB (Berlin Database of Emotional Speech)	German	5 emotions	535
SAVEE (Surrey Audio-Visual Expressed Emotion)	English	6 emotions	480

Work Division

ID	Task Name
1	EAMVCS
2	Literature Review
3	Dataset Collection & Preprocessing
4	Emotion & Speaker Feature Extraction
5	Multilingual TTS Integration
6	Testing & Evaluation
7	Documentation & Presentation



Software Hardware Requirements

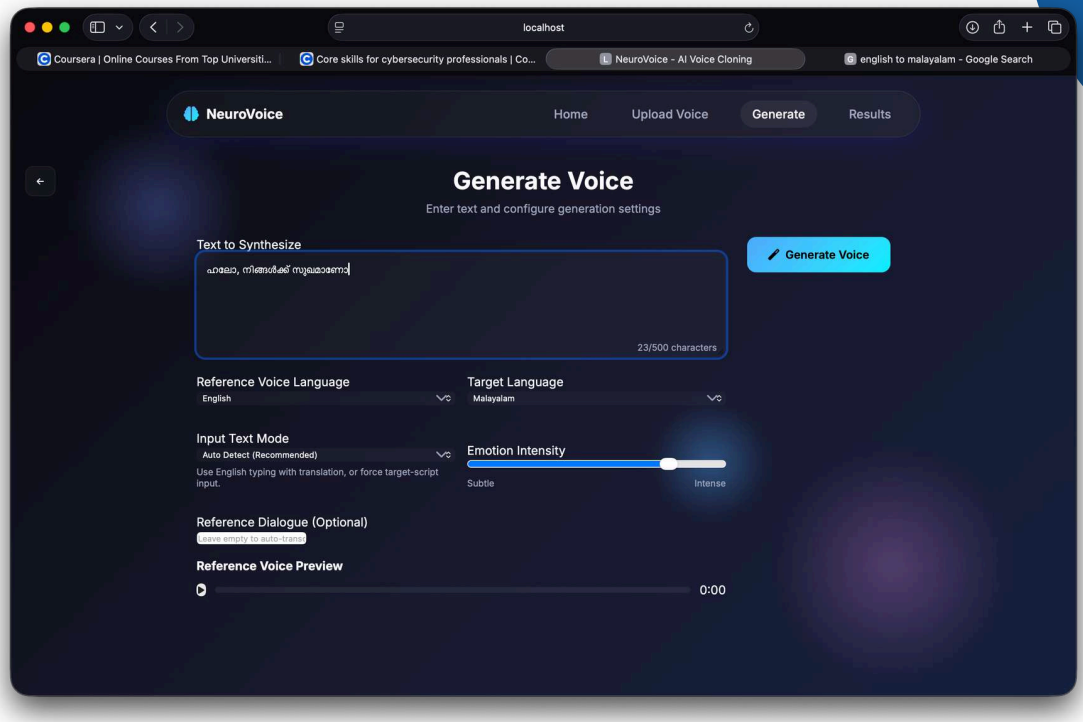
Hardware

- CPU: i5 or above
- RAM: 8–16 GB
- Storage: 30–50 GB
- GPU: NVIDIA GPU recommended (or Google Colab)
- Internet

Software

- OS: Windows / Mac
- Python 3.8+
- PyTorch + CUDA
- phonemizer / espeak-ng
- Resemblyzer / ECAPA-TDNN
- F5-TTS(VITS)
- HiFi-GAN (optional)

Results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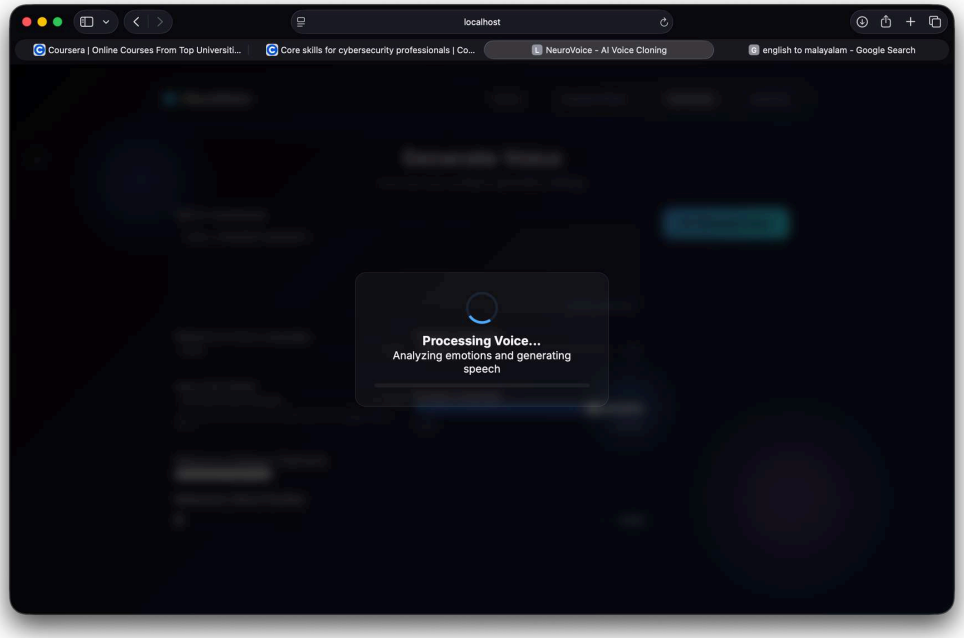
Coursera | Online Courses From Top Universiti...
Core skills for cybersecurity professionals | Co...
NeuroVoice - AI Voice Cloning
english to malayalam - Google Search
NeuroVoice
Home
Upload Voice
Generate
Results
Generate Voice
Enter text and configure generation settings
Text to Synthesize
ഹലോ, നിങ്ങൾക്ക് സുഖമാണോ?
23/500 characters
Generate Voice
Reference Voice Language
English
Target Language
Malayalam
Input Text Mode
Auto Detect (Recommended)
Use English typing with translation, or force target-script input.
Emotion Intensity
Subtle
Intense
Reference Dialogue (Optional)
Leave empty to auto-transc
Reference Voice Preview
0:00

Results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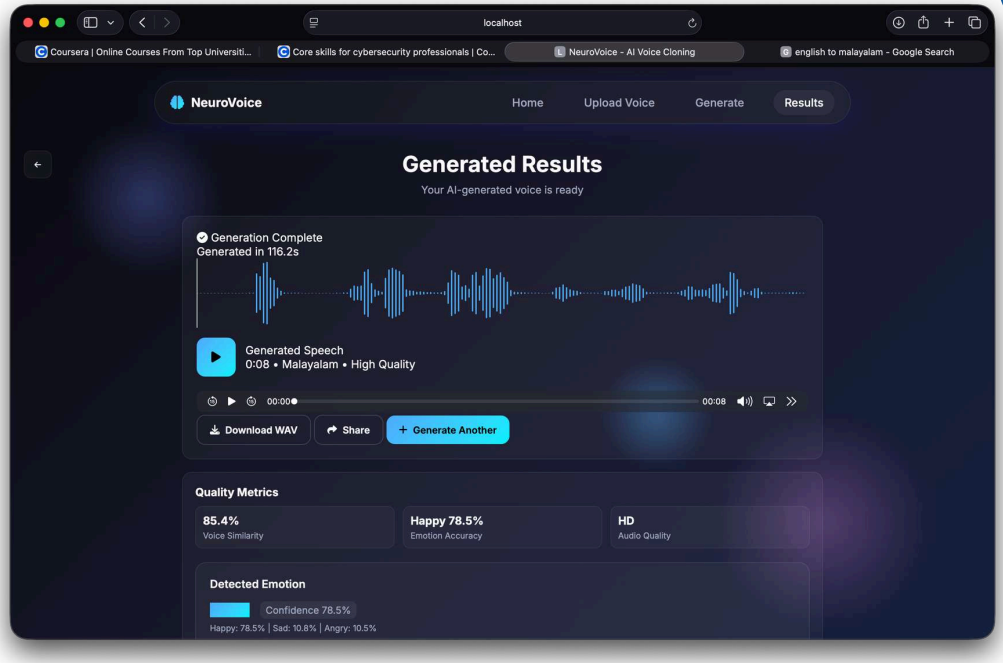
Coursera | Online Courses From Top Universiti...
Core skills for cybersecurity professionals | Co...
NeuroVoice - AI Voice Cloning
english to malayalam - Google Search
Processing Voice...
Analyzing emotions and generating speech

Results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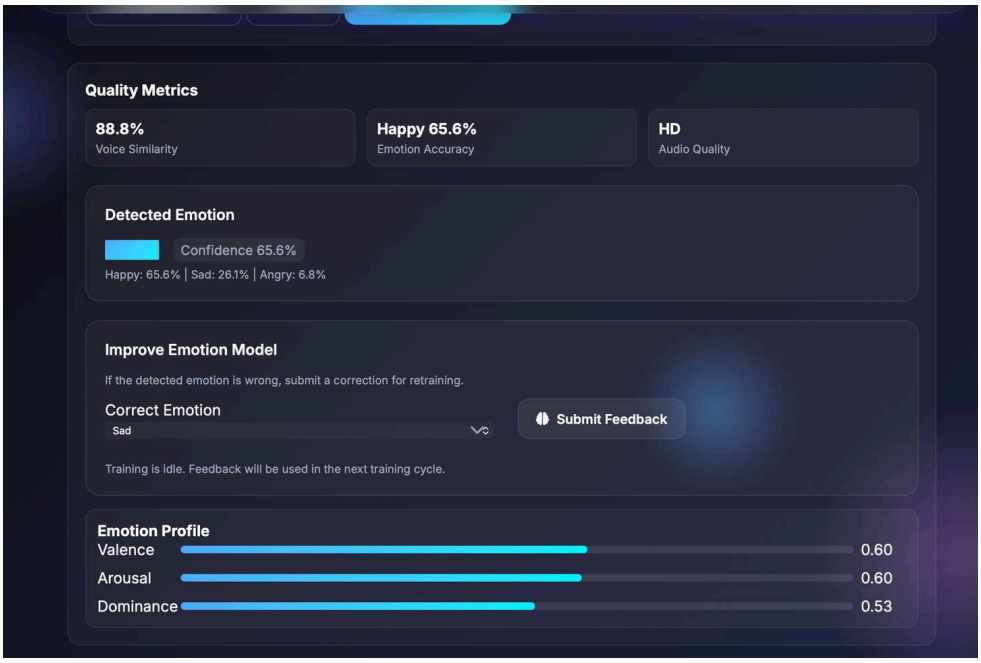
Core skills for cybersecurity professionals | Co...
NeuroVoice - AI Voice Cloning
english to malayalam - Google Search
NeuroVoice
Home
Upload Voice
Generate
Results
Generated Results
Your AI-generated voice is ready
Generation Complete
Generated in 116.2s
Generated Speech
0:08 • Malayalam • High Quality
00:00
00:08
Download WAV
Share
Generate Another
Quality Metrics
85.4%
Voice Similarity
Happy 78.5%
Emotion Accuracy
HD
Audio Quality
Detected Emotion
Confidence 78.5%
Happy: 78.5% | Sad: 10.8% | Angry: 10.5%

Results


Quality Metrics
88.8%
Voice Similarity
Happy 65.6%
Emotion Accuracy
HD
Audio Quality
Detected Emotion
Confidence 65.6%
Happy: 65.6% | Sad: 26.1% | Angry: 6.8%
Improve Emotion Model
If the detected emotion is wrong, submit a correction for retraining.
Correct Emotion
Sad
Submit Feedback
Training is idle. Feedback will be used in the next training cycle.
Emotion Profile
Valence
0.60
Arousal
0.60
Dominance
0.53

Conclusion

An Emotion-Aware Multilingual Voice Cloning System that generates natural speech in multiple languages while preserving the speaker's voice and emotional expression from a short reference audio clip.

The system extracts speaker identity and emotional prosody from a 5–10 second reference audio and combines them with multilingual text input to synthesize expressive speech.
It uses G2P phonemization, a YourTTS acoustic model, and a HiFi-GAN vocoder to generate high-quality multilingual speech that retains the original speaker's voice characteristics.

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